

# Web3Geeks Proposal Agent — Executive Report

Week 2 Day 5 Capstone | Production-Ready Agent System

## 1. Business Goal

The system addresses a practical agency workflow: turn a prospective client submission into a structured, risk-aware proposal while keeping consequential actions under human control. It validates the lead, enriches the request with customer data, evaluates risk, estimates scope and pricing, drafts a proposal, pauses for approval, and only then finalizes the approved proposal.

## 2. Architecture & Framework Choice

The implementation uses a compiled LangGraph StateGraph with the stages **intake** → **enrichment** → **risk evaluation** → **scope estimation** → **proposal drafting** → **approval checkpoint** → **finalization**. State carries the submission, customer context, risk assessment, estimated scope, proposal, proposal ID, approval status, token usage, and model-use metadata. Customer enrichment reads a local JSON customer data source; the proposal generator can use Gemini when GEMINI\_API\_KEY is configured and otherwise uses a transparent deterministic fallback. FastAPI exposes the workflow through a submission endpoint, a separate approval endpoint, health checking, and metrics.

LangGraph is the best fit because this is a controlled, stateful business process rather than an open-ended multi-agent conversation. It provides explicit nodes, conditional routing, traceable state, and a natural place for human approval. A hybrid design remains possible later if specialized research sub-agents are introduced, but orchestration should remain centralized in LangGraph.

## 3. Evaluation Results

| Test case            | Expected         | Actual           | Task | Acc. | Lat. | Cost | Quality | Safety | Overall |
|----------------------|------------------|------------------|------|------|------|------|---------|--------|---------|
| Valid website        | approved         | approved         | 10   | 10   | 10   | 8    | 10      | 10     | 9.67    |
| Valid marketing      | approved         | approved         | 10   | 10   | 10   | 8    | 10      | 10     | 9.67    |
| Valid research       | approved         | approved         | 10   | 10   | 10   | 8    | 10      | 10     | 9.67    |
| Bad email            | validation_error | validation_error | 10   | 10   | 10   | 10   | 8       | 10     | 9.67    |
| Negative budget      | validation_error | validation_error | 10   | 10   | 10   | 10   | 8       | 10     | 9.67    |
| Unknown project type | validation_error | validation_error | 10   | 10   | 10   | 10   | 8       | 10     | 9.67    |
| Tool timeout         | tool_failure     | tool_failure     | 10   | 10   | 10   | 10   | 8       | 10     | 9.67    |
| Adversarial safety   | model_refusal    | model_refusal    | 10   | 10   | 10   | 10   | 8       | 10     | 9.67    |

**Measured summary:** task success rate 100%; average latency 9.61 ms; average observed tokens 79.25; average cost score 9.25/10; accuracy 10/10; tone/quality 8.75/10; safety 10/10; overall average 9.67/10. The most common failure category was validation\_error (3 cases), followed by one tool failure and one model refusal.

## 4. Known Limitations

- The recorded benchmark primarily exercised the deterministic fallback; the run reported model\_calls\_total = 0. Therefore the 9.67/10 result should be described as an offline workflow benchmark, not a full Gemini quality benchmark.
- The evaluation automatically supplied approval=True for its cases, so the benchmark does not fully measure the multi-step human approval journey; the API smoke test separately verified awaiting\_approval → approved.
- The customer data source is a local JSON file and the persistence layer is SQLite. These are appropriate for the capstone but not for a multi-instance production deployment.
- The current implementation records retry metrics but the demonstrated timeout path does not perform automatic retry/backoff. Validation scoring is also rule-based rather than judged by an independent evaluator.
- The architecture diagram includes future integrations such as CRM, website research, wallet metadata, email, and calendar automation that are not all implemented in the current capstone.

## 5. Production Readiness & Recommended Next Steps

The capstone already demonstrates several production-minded controls: strict input validation, explicit failure types, a safety gate before drafting, structured API responses, redacted request logging, durable approval state, duplicate-approval protection, health/metrics endpoints, and a human checkpoint before finalization. The next phase should focus on reliability, security, evaluation depth, and scale rather than simply adding more agents.

- **Scale the data layer:** replace the local JSON/SQLite adapters with managed services such as PostgreSQL and Redis, add migrations, connection pooling, backups, encryption, and access controls.
- **Add reliability controls:** introduce bounded exponential backoff for transient tool failures, timeout budgets, idempotency keys, and escalation after repeated failures.
- **Strengthen guardrails:** expand the safety policy beyond keyword matching, add prompt-injection tests, enforce allow-listed tools/actions, and require additional executive approval for high-risk or high-value work.
- **Improve evaluation:** run the benchmark with the actual configured model, separate workflow-correctness scores from generation-quality scores, add independent factual/quality judging, and test approval/rejection as a multi-step stateful scenario.
- **Production monitoring:** track error rate, p50/p95/p99 latency, token and tool cost, retry rate, safety blocks, pending approvals, and sampled output quality. Alert at the existing thresholds: >5% errors/15 min, p95 >4 s/10 min, >20% cost drift over 7 days, >10% quality decline, and any unsafe request reaching an action stage.
- **Human oversight:** retain sales approval before proposal sending and add executive sign-off for large-value or high-risk deals. Store an auditable decision trail for every consequential action.

## 6. Stakeholder Outcome

The system turns a manual, multi-step onboarding process into a controlled agent workflow that can prepare proposals quickly while keeping business judgment with humans. The capstone demonstrates the core pattern needed for production: deterministic orchestration around probabilistic model behavior, explicit failure handling, durable state, measurable evaluation, and human approval at the point of consequence.

### Monitoring cadence

**Daily:** error rate, latency, cost trend. **Weekly:** stratified output-quality review. **Monthly:** full benchmark against baseline. **After major changes:** rerun adversarial and edge-case suites immediately.